Programmable 4-Wheel Toy Car Control Circuit Package

A beginner-friendly programmable car using two driven rear wheels, two free-rolling front wheels, an ESP32 controller, and a TB6612FNG dual motor driver.

Assumptions

• Drive layout assumed: 2WD differential drive with one DC gear motor on the left side and one on the right side; front wheels are passive.

• Programming platform chosen: Arduino-compatible ESP32 dev board because it is easy to program and has enough PWM-capable GPIO for motor control.

• Battery assumed: 2-cell 18650 holder in series for about 7.4 V nominal, 8.4 V fully charged.

• Motor type assumed: common TT-style brushed DC gear motors rated for about 3-6 V operation.

• This is a concept/prototyping package, not a production automotive or toy safety certification package.

• Links are representative product pages from common vendors or manufacturers; live stock and exact price were not independently verified at delivery time.

Basic circuit description

• The battery pack feeds a master power switch. After the switch, the raw battery line powers the motor supply input of the TB6612FNG motor driver.

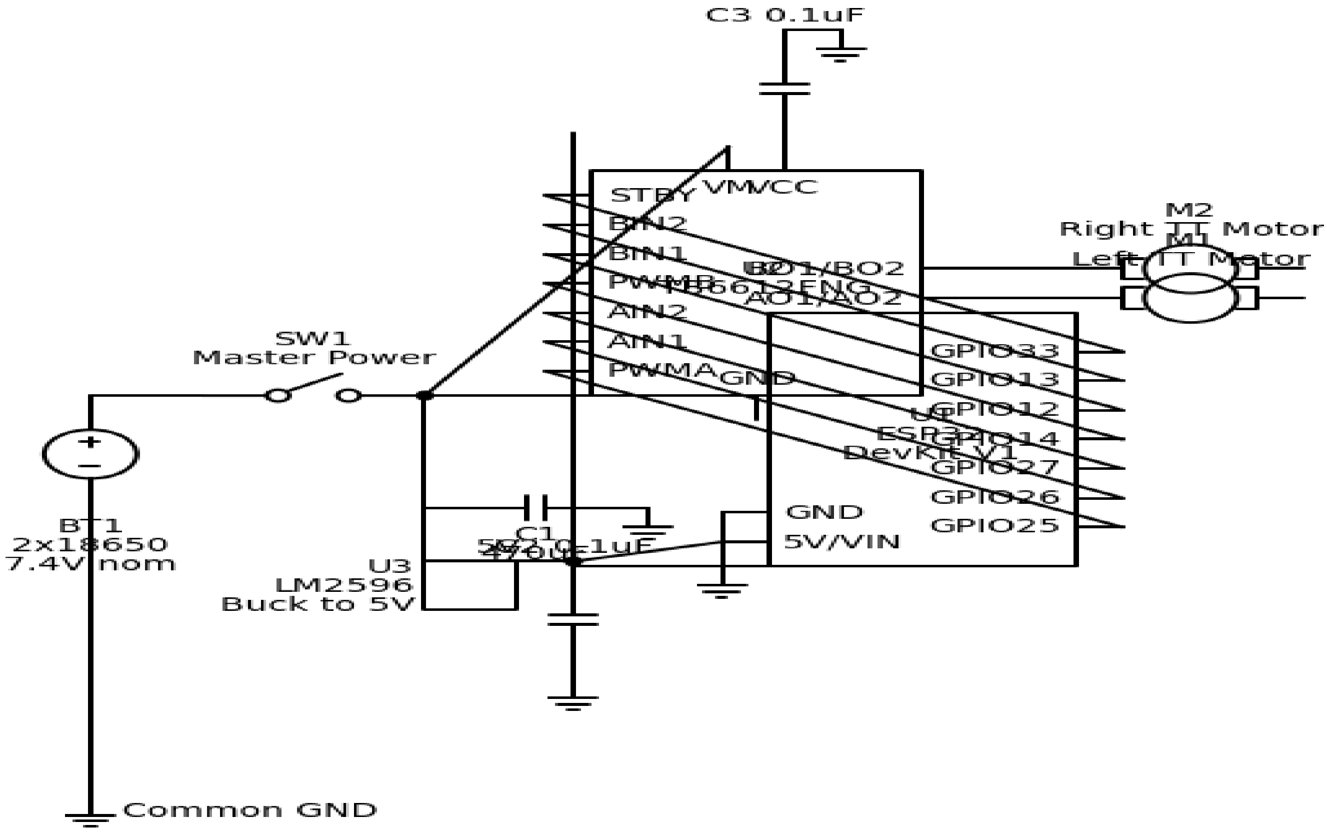
• The same switched battery line also feeds an LM2596 buck converter adjusted to 5 V. That regulated 5 V rail powers the ESP32 dev board through its 5V/VIN input and powers the TB6612FNG logic VCC pin.

• The ESP32 drives PWMA, AIN1, AIN2, PWMB, BIN1, BIN2, and STBY on the TB6612FNG. This allows independent speed and direction control of the left and right motors.

• Each motor output from the TB6612FNG connects to one TT motor. Grounds for battery, buck converter, motor driver, and ESP32 are all tied together.

• Bulk and decoupling capacitors are included near the motor supply and logic rail to reduce resets and noise from the brushed motors.

Schematic



Suggested firmware pin map

ESP32 pin	TB6612FNG pin	Purpose
GPIO25	PWMA	Left motor PWM
GPIO26	AIN1	Left motor direction 1
GPIO27	AIN2	Left motor direction 2
GPIO14	PWMB	Right motor PWM
GPIO12	BIN1	Right motor direction 1
GPIO13	BIN2	Right motor direction 2
GPIO33	STBY	Driver standby enable

Bill of materials

Item	Refs	Qty	Description	MPN / supplier	Purchase link
1	U1	1	**ESP32 DevKit V1 development board** ESP32-WROOM-32, Arduino-compatible, 30-pin dev board, Wi-Fi/BLE, 5 V input via VIN/5V, 3.3 V logic Any widely used ESP32 DevKit with exposed GPIO and VIN/5V pin is acceptable; confirm actual pin labels before wiring.	ESP32 DevKit V1 (generic CP2102 type) Adafruit / generic distributors	product page
2	U2	1	**Dual DC motor driver breakout** TB6612FNG, 4.5-13.5 V motor supply, 2.7-5.5 V logic, about 1 A continuous/channel Chosen over L298N because it is more efficient and better suited to battery-powered toy cars.	Pololu 713 TB6612FNG carrier Pololu	product page
3	M1,M2	2	**TT style brushed DC gear motor** 3-6 V rated, about 160-200 RPM at 6 V, brushed motor for small robot car If your selected motors have much higher stall current, upgrade the driver and power wiring.	Adafruit 3777 or equivalent Adafruit	product page
4	U3	1	**Buck converter module** LM2596 adjustable step-down module, input >5 V, output set to 5.0 V, up to about 2 A practical without extra cooling Adjust output to 5.0 V with a multimeter before connecting the ESP32.	LM2596 module SunFounder / generic distributors	product page
5	BT1	1	**2x18650 battery holder with switch** Series holder, nominal 7.4 V output, wire leads, integrated on/off switch If your holder already includes a switch, the separate SW1 item can be omitted.	2-cell 18650 holder with switch Seeed Studio	product page
6	B1,B2	2	**Protected 18650 Li-ion cells** 3.6-3.7 V nominal, around 3200-3400 mAh, protected preferred for beginner use Use a proper Li-ion charger. Do not charge cells while wired into an improvised circuit unless a correct charging/protection design is added.	Panasonic NCR18650B protected BatteryTekUSA	product page
7	SW1	1	**Master power switch** SPST on/off, low-voltage DC switching, panel-mount or inline Optional if battery holder already includes adequate switch and current handling.	C&K; DA series SPST rocker or equivalent C&K;	product page
8	C1	1	**Bulk electrolytic capacitor** 470 uF, 16 V minimum, radial, low-ESR preferred Place across VM and GND near the motor driver.	Panasonic EEUFR1C471 RS	product page
9	C2,C3,C4	3	**Ceramic decoupling capacitor** 0.1 uF, 50 V, radial or leaded ceramic Use near ESP32 supply and motor driver logic supply.	Jameco 15270 or equivalent Jameco	product page
10	J1,J2	2	**2-pin screw terminal block** 5.08 mm pitch, >=10 A preferred for battery/motor wiring Optional when modules already provide terminals or prewired leads.	Weidmuller 1760490000 or equivalent RS	product page
11	HW1	1	**4-wheel robot chassis kit** Chassis with 4 wheels and mounting hardware, compatible with TT motors Mechanical item included for completeness; exact chassis can vary.	Generic 4-wheel robot chassis Generic robotics supplier	product page

Limitations and next steps

• Add wheel encoders, an ultrasonic sensor, or line sensors if you want autonomous navigation rather than timed open-loop driving.

• Before connecting the ESP32, verify the buck converter output with a meter and confirm battery polarity.

• Brushed motor noise can reset microcontrollers; keep power wiring short and consider additional suppression capacitors directly on the motors if needed.

• If you expect motor stall currents above about 1 A continuous per channel, select a higher-current driver and heavier-gauge wiring.